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Adverse impacts of metaverse-induced cognitive biases on the immersive shopping experience: A conceptual model developed from a qualitative approach

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ABSTRACT

Immersive shopping (IS) is the usage of simulation-based technology like metaverse to create interactive and highly personalised shopping experiences for customers wherein they spend quality time selecting the products, which increases their familiarity with the brand. When compared to a standard shopping experience, the IS experience (ISE) offers brick-and-mortar retailers a positive brand image. However, the metaverse can induce cognitive biases (CBs) in customers that can negatively influence their reasoning and decision-making. CB is the systematic error in thinking that occurs when people are processing and interpreting information during the shopping, which affects their decisions and judgments. Therefore, this study explores the adverse impacts of metaverse-induced CBs on ISE for brick-and-mortar retail customers. Using simple random sampling, data were collected from 20 customers, and a qualitative approach was used for data analysis. The finding produces three adverse impacts for ISE. First, metaverse-induced CBs create a digital divide between customer communities, and the integration of retail services with the metaverse further aggravates the risk of this divide. Second, metaverse-induced CBs create financial malfeasance, which makes the metaverse susceptible to financial biases. Third, metaverse-induced CBs increase business reputation risk by adversely impacting decision-making, strategy formulations, and outcomes.

1. Introduction

A metaverse is an immersive, 3D virtual space (3D VS) where users can engage with one another through avatars for daily interactions, unlocking the possibility for communication, trade, and new opportunities on a global scale (Rehman et al., 2024) (Ki et al., 2025). Artificial intelligence (AI), digital twins, augmented reality (AR), virtual reality (VR), blockchain, cloud computing, and edge computing are among the cutting-edge technologies that are converging in the metaverse (Al-Emran and Deveci, 2024) (Abdulmuhsin et al., 2024). A technology is deemed immersive if it creates a realistic experience that successfully immerses the user in the simulated environment (Hadi et al., 2024) (Alkhwaldi, 2024a) (Alkhwaldi, 2024b). Hence, the immersive aspect of the metaverse hinges on its potential for broad strategic and significant positive effects (Marabelli and Newell, 2023) (Balhareth et al., 2024) (Jafar et al., 2024). Additionally, businesses have started investigating the metaverse as a business platform in an effort to take advantage of its possibilities (Park, 2024), despite the fact that there is not yet a clear consensus on its merits (Park and Lim, 2023). Additionally, businesses assess the potential of metaverse and look for ways to adopt it into their current business models (Büchel and Spinler, 2024). In fact, retail has taken notice of the growing interest in the metaverse in recent years (Luong et al., 2024).

In continuation, the metaverse has relevance in the retail i.e., it is anticipated that the metaverse can significantly change how consumers interact with the digital environment and change the retail (Yoo et al., 2023). Retail businesses are using the metaverse to spread the word about their marketing plans, mostly via providing digital products and immersive brand experiences (Park and Lim, 2023). The metaverse expands educational opportunities with virtual classrooms and

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complements physical retail places with virtual shopping (Yaqoob et al., 2023). With the immersive 3D environments and distinctive customer experiences, the metaverse is slowly overtaking traditional retail environments (Klaus and Manthiou, 2024). Retail brands like Nike, Puma, Gap, Clarks, Tommy Hilfiger, and Gucci have started to look into ways to interact with customers in the metaverse (Barrera and Shah, 2023). In the current environment, digitalisation uses the metaverse, which benefits retail businesses by reducing costs, providing services that can be personalised and increasing customer satisfaction (Kar and Varsha, 2023). The advent of the metaverse has offered new avenues for people to reach the digital world, expanding the product and service market for retailers (Xiang et al., 2024).

However, the brick-and-mortar retail stores rooted in the metaverse can have adverse effects on customer consumerism and behaviour. In consumerism, a customer's thought process can be dangerously deviated from rationality by changing the way they view a retailer's products, services, decision-making, or processes rationally by convincing them to have pre-existing and unchallenged notions about the same. The negative behaviours of customers may include returning products, cancelling orders, or a range of other actions. These apparent shortcomings take shape due to cognitive biases (CBs) that can be present in a customer's perception due to metaverse-induced information. A CB is an irrational conclusion or view that people form as a result of inaccurate factual reporting or inaccurate decision-making heuristics (Shuraida and Titah, 2023). Customers' cognitive abilities may be influenced by CBs during the decision-making process, which could result in systematic errors (Athota et al., 2023). Therefore, metaverse-induced CBs are systematic errors in customers thinking that occur when they receive and interpret information from the metaverse and have a negative impact on their ability to make decisions and judgements. However, it is crucial to understand that CBs can have negative impacts on a customer's thinking in immersive shopping (IS).

IS involves the usage of AR, and VR technologies to create a more engaging and interactive shopping experience, and features like 3D product displays, virtual try-on capabilities, and personalised recommendations can be part of this. Compared to traditional online retail stores, the 360-virtual store represents a more realistic and IS environment (Ruusunen et al., 2023). Brand managers of a retail firm that sells through their directly owned store may plan to integrate AI-powered VR and AR technology that might give customers IS experiences (ISE) (Sestino and Amatulli, 2024). ISE is a synthetic, totally digital, or digitally enhanced environment where customers can experience sensory presence made possible by immersive technology during their shopping process. In fact, AR can enhance customers' shopping experiences by integrating product displays into real-world environments and providing them additional information about such products (Lin and Huang, 2024).

Literature has discussed CBs in the retail, i.e., heuristic bias and conservative bias are two CBs that have a significant impact on the pricing effect of historical stock downside risks in the retail stock markets (Li et al., 2023). High-passion customers are able to sift through information to use for in-store evaluation and to minimise the effects of CBs (Kumar, 2023). Literature has discussed ISE in the retail, i.e., AR emerges as a potent technology that promotes customer satisfaction and loyalty in the retail industry due to its capacity to provide an engaging ISE (Sharma et al., 2023). Retailers can develop an ISE using image interactivity technology that reduces the requirement for physical touch, enhancing the effectiveness of online shopping (Cano et al., 2017). Based on the above arguments, there is a dearth of literature that has (1) presented theoretical background on ISE, metaverse-induced CBs, and negativity bias; and (2) investigated the lack of knowledge between metaverse-induced CBs and negativity bias and subsequently between negativity bias and ISE all in brick-and-mortar retail.

Moreover, the negativity bias is the tendency of customers to give greater importance to negative experiences in IS than to positive or neutral experiences. Following the gaps, the literature has indicated future research into CBs and ISE. Future research is required to understand how CBs influence professionals in the retail (Athota et al., 2023). Future research should examine technology-induced CBs (Shuraida and Titah, 2023), and the metaverse as an immersive technology is no exception. Future research should focus on the effect of new emerging technologies on ISE for customer decision-making (Sharma et al., 2023). Future research should merge emerging technologies to deliver ISE that frequently locks in customers (Chen et al., 2018). Regarding research problematisation, disquisition on ISE and CBs is paramount in the current era, i.e., ISE offers customers novel ways for interacting electronically with retail products (Chen et al., 2022). ISE can boost retail revenues and attract new customers (Butt et al., 2023). More ISE benefits retailers (Hsu et al., 2024). CBs can be the barrier to the success of an organisation (McLaren et al., 2025). CBs have the potential to intensify barriers to communication and intergroup conflicts in online media (Huang et al., 2025). Therefore, the gaps in the literature, the necessity of future research, and the research problematisation justify the convincing motivation.

The scope of the study is to explore the adverse impacts of metaverse-induced CBs on ISE for brick-and-mortar retail customers by developing a conceptual model from a qualitative approach. To achieve it, the contributions of the study are to (1) provide theoretical background on ISE, metaverse-induced CBs, and negativity bias; and (2) obtain insights between metaverse-induced CBs and negativity bias, negativity bias and ISE. The novelty of the study is to (1) explore the occurrence of CBs due to metaverse that skews ISE; and (2) contribute to the theory by proposing a new conceptual model wherein metaverse-induced CBs are the independent variables, negativity bias as an intermediate variable, and ISE as a dependent variable. In addition, this study strongly argues that (1) CBs are explored in diversified areas (e.g., psychology, sociology, finance, etc.) but not on metaverse, which can introduce a new metaphor; (2) metaverse-induced CBs can lead to distorted thinking, and its impact on brick-and-mortar retail customers can be large as it hinders decision-making and can cause customers to misjudge risks and threats; and (3) ISE can propel brick-and-mortar retail customers to embrace the metaverse in shopping. Therefore, this study strongly argues for its worthiness.

AI and the metaverse are two examples of complementary but distinctly different information technology (IT). The former facilitates better decision-making, while the latter provides immersive experiences and activities. Users in metaverse-based systems can interact with each other in their digital environments, creating a whole new parallel universe that transcends physical boundaries. AI-based systems can learn and adapt to the data they are exposed to. Thus, metaverse-induced CBs differ from AI-induced CBs in that the former affect user experiences and interactions, whilst the latter affect outputs and decision-making. The metaverse is a persistent, immersive, and networked digital world where users interact through avatars, and AI focuses on making machines capable of activities that normally require human intelligence. Therefore, AI-induced CBs appear in the outputs and decisions made by AI systems, whereas metaverse-induced CBs are mostly based on the perceptions, assessments, and actions of human users within the 3D VS.

The following is the structure of the remaining sections: In Section 2, the theoretical background is covered. In Section 3, the research methodology is presented. This comes after Section 4's results and Section 5's discussion. Finally, the study concludes in Section 6.

2. Theoretical background

CBs imply forces that frequently affect an individual's thought process without their knowledge, making them difficult to identify and manage (Kunkler and Roy, 2023). Despite the possibility that some CBs evolved as expedients for productive action, others may be the result of cognitive constraints and restricted rationality (Ciriello and Loss, 2023). Therefore, it can be concluded that CBs have various consequences for IS equilibrium. However, customers explore and interact with products

using IS in a virtual environment, where the metaverse provides new opportunities for collaboration that enable geographically dispersed teams to work together virtually (Richter and Richter, 2023). Immersive experiences provide customers the chance to connect with products and services in a realistic virtual setting, improving engagement, providing personalised experiences, and enabling direct and enlightening feedback, and all of which contribute to the development of stronger customer relationships (Hajian et al., 2024). In fact, the demands of Generation Z for immersive offers of products and services are growing (Schiopu et al., 2022). Therefore, the above discussion justifies the need for theoretical background, which covers ISE, metaverse-induced CBs, and negativity bias.

2.1. Immersive shopping experience

Customer expectations for brick-and-mortar retail stores were irrevocably changed by online shopping, and now immersive shopping is reshaping the retail industry. Instead of rapid navigation, IS aims to provide customers with an interactive and engaging experience that makes them feel really involved in the shopping session. A new virtual environment called the metaverse allows users to inhabit and interact with one another in immersive environment (Ahn et al., 2024). A person's immersive experience in the 3D VS is improved by the core technologies of the metaverse (Cheng et al., 2022). Metaverse describes an immersive environment that exists outside of the natural universe with unlimited possibilities (Gleim et al., 2024). Therefore, the ISE is an online shopping experience that makes use of simulation-based software like VR, AR, machine learning (ML), AI, and chatbots to provide a seamless, interactive, and comfortable environment to customers in the metaverse. Since many customers lead busy lives and have limited time for buying, immersive commerce offers a convenient, engaging alternative to shopping online with an ISE (Zhao et al., 2022). Moreover, ISE positively impacts purchase intention and actual sales (Zimmermann et al., 2023).

In continuation, brick-and-mortar retailers may create a deep emotional connection with a customer and transform them into a devoted supporter by employing metaverse to create an entertaining and memorable ISE (Kim et al., 2023). Moreover, the ability of the metaverse to offer customers a plethora of timely, personally relevant information as well as a highly personalised, immersive, and interactive support system for organising and managing their experiences has ramifications for how they make decisions (Gursoy et al., 2023). Therefore, ISE is a multisensory one that is made possible by a number of user interactions that give the customers intuitive and emotional value, and different formats can be used to present these interactions, i.e., (1) *flat interfaces:* with the use of interactive touch screens; (2) *natural interfaces:* voice, spatial, gesture-based, auditory, or olfactory; and (3) *extended reality interfaces:* 3D, augmented, virtual, or mixed. When compared to a traditional shopping experience, ISE offers retailers distinct advantages, including an accelerated decision to buy, more personalised experiences, improved customer engagement, improved marketing campaigns, boosted conversion rates, a lower number of returns, and more consumer loyalty.

For a better ISE, the retail industry must make the public aware of its virtual products and virtual services. In recent years, the rise of immersive technologies has led to a significant change in the marketing landscape (Wang et al., 2023). Therefore, VR, AR, and other interactive technologies are all part of immersive marketing, which has become a revolutionary method of engaging customers (Bilgihan et al., 2024). Hence, immersive marketing can shape the future of ISE by (1) *advertising promotion:* virtual brands can create engaging and dynamic advertising experiences with immersive marketing by integrating with metaverse platforms. Therefore, marketers may maximise their efforts in search engine optimisation (SEO), and advertising by utilising metaverse; (2) *content experience:* customers can have immersive, personalised storytelling experiences with metaverse tools used for content creation, curation, and dissemination. Therefore, by combining these tools with SEO tactics, marketers can redefine customer connection and encourage greater brand loyalty. Hence, virtual brands may create strong, enduring relationships with customers through ISE, which can raise awareness, engagement, and loyalty of such brands.

Moreover, individual characteristics such as priorities, preferences, morals, and values greatly influence consumer behaviour, particularly in ISE. According to Gallin and Portes (2024), ISE allows customers and businesses to communicate in real time, and one of the main factors influencing consumer behaviour is communication. Customers experience an ISE as a result of promotional activities (Zhang and Li, 2024), and these tactics have a big influence on consumer behaviour in the retail industry. ISE can increase the like hood of impulse purchases (Moghddam et al., 2024), which has been regarded as a type of consumer shopping behaviour (Nyrhinen et al., 2024). However, immersive shopping is an essential component of any retail strategy since it can allow building imaginative virtual experiences that can mimic or replace the in-store experience, which plays an important factor behind its growth of consumer behaviour (Erensoy et al., 2024). Understanding consumer behaviours and the underlying motivations behind them is crucial to maximising the potential of immersive virtual reality technologies in the retail industry. Therefore, the above discussion signifies that ISE can improve the consumer behaviour in the retail.

2.2. Metaverse-induced cognitive biases

A CB is a methodical, foreseeable, and repeatable trend that can challenge information systems that presumptively have reasonable use intentions (Ciriello and Loss, 2023), which usually occurs while making incorrect conclusions (Athota et al., 2023). Due to CBs, it can be challenging for businesses to distinguish between short-term and long-term decision-making processes (Shen et al., 2023). Therefore, decision-making and overall outcomes in shopping can be influenced by CBs in today's rapidly advancing technological landscape. In fact, in the context of metaverse, these CBs can lead to unintended consequences on IS. The pervasive influence of CBs in technology has been discussed in literature. Hence, with consideration of the above discussions, this study proposes eight CBs that can be induced by the metaverse, which include confirmation bias, availability bias, anchoring bias, egocentric bias, hindsight bias, self-serving bias, choice-supportive bias, and conformity bias. The rationales behind consideration of eight CBs are justified, i.e., confirmation bias influences the selection of information (Lee et al., 2024), which can cause the metaverse system to provide users the information that reinforces their existing viewpoints in IS.

In continuation, information system complexity and a lack of knowledge about which data objects to integrate can lead to availability bias (Vincent et al., 2020), which can cause the metaverse system to provide users with the information that makes their judgements based on easily available information, rather than what is most accurate or relevant in IS. An important bias that manifests itself in market analysis is anchoring bias (Bonaccorsi et al., 2020), which can cause the metaverse system to provide users the information that enables them to rely heavily on the first piece of information they encounter when making decisions in IS. In the presence of a concurrent task, the egocentric bias increases (Eligio et al., 2012), which can cause the metaverse system to provide users the information that causes them to rely too heavily on their own point of view higher than that of others in IS. Due to the distortion of recalled information, hindsight bias has a predictable impact on an individual's nascent experience (Cassar and Craig, 2009), which can cause the metaverse system to provide users the information that causes them to convince themselves after an event that they accurately predicted it before it happened in IS. Managers must understand the relationship between various forms of customer autonomy and self-serving bias while using co-creation approaches (Zhu et al., 2024), which can cause the metaverse system to provide users the information that attributes successes to their internal factors (e.g., their own

abilities) and failures to external factors (e.g., bad luck or others) in IS.

Moreover, with the use of online social networks, choice-supportive bias plays a role in decision-making (Sadovykh et al., 2015), which can cause the metaverse system to provide users the information that tends to make them remember their choices as better than they actually were, attributing more positive and fewer negative aspects to them, while doing the opposite for rejected options in IS. After viewing other users' historical ratings, people exhibit conformity bias (Pan et al., 2024), which can cause the metaverse system to provide users the information that develops the tendency to change their beliefs or behaviour to fit in with others in IS. The study has not considered other CBs like the Recency effect and the Dunning-Kruger effect. The Recency effect is a memory phenomenon in which customers tend to recall the most recent metaverse-induced information more accurately (Kim and Sanders, 2002), but it is most suited to strategic options. In the Dunning-Kruger effect, a customer with low metaverse competency in a retail domain tends to inflate their own abilities, which is not applicable in this study because (Bahreini et al., 2023) it looks at how one evaluates oneself against oneself as well as oneself against others. CBs undermine decision effectiveness (Cabitza et al., 2023), and hamper purchase decisions (Ressel et al., 2024). So, the scope of this study is limited to the adverse impact of them. Each CB is further elaborated.

2.2.1. Confirmation bias

It is the tendency of customers to deliberately seek out and select information that confirms their pre-existing beliefs about a brick-and-mortar retailer's products, services, decision-making, or processes (PSDP) induced in the metaverse while disregarding or downplaying contradictory evidence. In essence, it is a propensity that can affect a customer's deeply held beliefs, retail politics, and personal decisions. Confirmation bias has been linked to different information-processing philosophies among individuals (Alsaad et al., 2018). Therefore, customers' confidence and sense of validation are increased when they encounter metaverse-induced information that supports their pre-conceptions in IS. In contrast, customers could feel frightened and go through cognitive dissonance when they encounter metaverse-induced information that goes against their beliefs in IS. Consequently, they might then dismiss or rationalise such information about IS. Confirmation bias cannot be totally eliminated by the metaverse, but by being aware of it and making conscious efforts to control it, the effects can be mitigated. Customers can learn methods to analyse information objectively and become more aware of such biases by receiving education and training in critical thinking techniques.

2.2.2. Availability bias

It is the tendency of customers to over-anticipate or over-estimate their experiences with a brick-and-mortar retailer's metaverse-induced PSDP, which has greater availability in recent times to influence their sentiments, intentions, or satisfactions despite a lower probability of the same experience being repeated. In essence, it is a propensity in which a customer makes a decision depending on how quickly recent, frequent, extreme, vivid, and negative analogous examples or situations come to mind. Snap judgements are influenced by availability bias, which means that instead of carefully considering all the facts and better options, one takes a streamlined approach that concentrates on the options that immediately spring to mind (Elston, 2023). Therefore, customers frequently favour recent metaverse-induced information when making judgements in comparison to recent news or experiences, which are easier to recall and frequently shape and distort fresh opinions or events in IS. Availability bias cannot be totally eliminated by the metaverse, so customers must be aware of this and make an effort to evaluate information and options in a more comprehensive and balanced manner when making decisions or forming opinions.

2.2.3. Anchoring bias

It is the tendency of customers to rely heavily on an anchor (i.e., a specific piece of information or reference point) in a brick-and-mortar retailer's metaverse-induced PSDP while making consequential decisions. In essence, it is a propensity in which a customer is presented with information that influences their subsequent thinking and evaluation of a situation. In anchoring bias, individuals frequently base their current opinions on conspicuous facts, which cause them to incorporate new information more slowly (Hao et al., 2024). Therefore, once an anchor is set, it acts as a mental point of comparison that affects the customer's subsequent decisions in IS. So, instead of starting over or evaluating all the material objectively, the customers base their judgements or estimates on the original anchor. Negotiations, price plans, and subjective assessments are just a few of the decision-making processes that might be impacted by the anchoring effect in IS. It can also result in the overvaluation or undervaluation of products, services, or other types of information, depending on the initial anchor that is offered. Anchoring bias cannot be totally eliminated by the metaverse, so customers can weigh different sources of information and critically assess the original anchor's relevance and veracity by being aware of the anchoring effect.

2.2.4. Egocentric bias

It is the customer's overconfidence or subjective confidence in their judgements on a brick-and-mortar retailer's metaverse-induced PSDP whose accuracy is challenged. In essence, it is a propensity in which a customer demonstrates overconfidence in three ways i.e., (1) Overestimation of actual performance, in which customers expect themselves to perform or produce better results than their real skills or past performance suggest. They could overestimate their abilities, expertise, or skills for a certain task. (2) Overestimation of performance relative to others, in which customers overestimate their performance or abilities compared to others. They believe they are more competent, skilled, or superior to their colleagues or rivals, despite the fact that objective measurements may indicate otherwise. (3) Overconfidence in expressing unwarranted certainty about beliefs, in which customers display unjustified complete trust in the accuracy or validity of their beliefs, opinions, or judgements. Even when there is scant or conflicting data to support their views, they may nonetheless express strong opinions or steadfast certainty.

In continuation, when estimating how others feel in response to stimuli, individuals frequently exhibit an egocentric bias and overestimate how much other individuals would feel the same way (Reshadi, 2023). Therefore, these multiple instances of overconfidence demonstrate how ubiquitous this bias is and how it affects judgement, problem-solving, and interpersonal relationships in IS. The overconfidence effect can also cause customers to predict the future with excessive optimism, take unnecessary risks, ignore opposing points of view, and pass judgement incorrectly in IS. Egocentric bias cannot be totally eliminated by the metaverse, so customers can lessen the effects of overconfidence and make more informed and balanced decisions by developing awareness of their limitations, seeking out different viewpoints, and establishing a healthy scepticism towards their own judgements.

2.2.5. Hindsight bias

It is a subtle feeling of determinism about a brick-and-mortar retailer's metaverse-induced PSDP that leads a customer to believe that the predictability of past events is higher than it actually is, with greater certainty. In essence, it is a propensity in which a customer demonstrates overconfidence in predicting future events. Managers may not be able to see strategic opportunities due to hindsight bias (Biedenbach et al., 2015). Therefore, hindsight bias can have significant effects on IS. More broadly, this bias prevents customers from learning from their experiences. It can cause customers to overestimate their ability to see the future and make decisions based on metaverse-induced information, which might impair their judgement and decision-making in the future. Hindsight bias cannot be totally eliminated by the metaverse, so to

reduce its impacts and encourage more accurate evaluations of the past and future forecasts, it is essential for customers to recognise and understand this bias in IS.

2.2.6. Self-serving bias

It is the tendency of customers to take greater credit for successful acceptance or adoption of a brick-and-mortar retailer's metaverse-induced PSDP than failures. In essence, it is a propensity in which a customer holds a positive opinion of themselves, motivated by the desire to uphold and enhance their self-esteem. Working with a machine that has a different personality from their own prevented individual results in self-serving bias (Wang et al., 2022). Therefore, self-serving bias can cause customers to have an unduly favourable self-perception of IS. It entails attributing accomplishments to one's skills and efforts and failures to uncontrollable outside events. Customers defend their self-esteem against danger and harm by rejecting unfavourable criticism, highlighting their accomplishments and strengths, and downplaying their flaws and failings. Illusions and misconceptions are created and spread by these cognitive and perceptual tendencies. Self-serving bias cannot be totally eliminated by the metaverse. So to reduce its impacts, customers must not make quick decisions, must make an effort to give others credit, practice self-compassion, see the opportunity in mistakes, and consider all contributing factors.

2.2.7. Choice-supportive bias

It is the customer's tendency to emphasise the positive aspects of a brick-and-mortar retailer's metaverse-induced PSDP according to their choice while downplaying the negative aspects of their choice. In essence, it is the propensity with which a customer has made a decision and how it can influence others to perceive and remember their choices. Choice-supportive bias reflects a tendency in which, after making a choice, individuals prefer to think that their choice was indeed the best one, which makes them less likely to change their minds afterwards (Shuraida and Titah, 2023). Therefore, a customer's perspective on their decision and future decision-making processes may be impacted by choice-supportive bias in IS. Customers may be more inclined to select their initial choice in similar circumstances by emphasising its benefits, even when objectively better alternatives are available. Choice-supportive bias cannot be totally eliminated by the metaverse, and being aware of it can help customers approach decision-making more objectively. Customers should strive for more balanced evaluations and reduce the influence of this bias on their decision-making outcomes by carefully examining the advantages and disadvantages of various solutions.

2.2.8. Conformity bias

The tendency of a customer to inherently follow and act according to a majority of certain beliefs and characteristics about a brick-and-mortar retailer's metaverse-induced PSDP for socialisation and conforming to the normality of a larger group. In essence, it is a propensity in which a customer spreads specific actions and beliefs that can influence public opinion or behaviour. Conformity bias affects how individuals evaluate polarised information (Mirhoseini et al., 2023). Therefore, it demonstrates how beliefs, concepts, fads, and trends grow in strength and acceptance as more individuals participate in IS. It has its roots in an individual's propensity to seek out social approval and follow accepted rules. Customers frequently seek guidance from others on how to act or what to believe, particularly when they notice a sizable number of other people adopting that behaviour or sharing that view. This may have a domino effect, in which the adoption of particular attitudes or behaviours spreads quickly throughout social networks. Conformity bias cannot be totally eliminated by the metaverse, so customers can maintain critical thinking, take into account other viewpoints, and make well-informed judgements based on their values, knowledge, and assessment of the available facts by being aware of its effect.

2.3. Negativity bias

It is a bias wherein a customer is strongly influenced or motivated by negative retail experiences or stimuli rather than positive ones, which can cause an unwanted disdain for the metaverse. In essence, it is a propensity that affects a customer's feelings, thoughts, actions, and can have less-than-desirable effects on decision-making. However, it is important to note that the negativity bias does not suggest that customers have just negative experiences or that good things do not matter for the metaverse. Instead, specific unpleasant events have an impact on customer memory, beliefs, and emotions that is more pronounced and permanent. When applied to customer choices to embrace novel technologies, negativity bias may create a psychological barrier that adversely affects customer perceptions of the intrinsically contradictory characteristics of the novel technology and, as a result, influences less favourable adoption choices (Kagan et al., 2023).

In continuation, owing to negative bias, negative messages are given greater weight by people than positive ones, and as a result, individuals may be more likely to believe those who convey negative sentiment (Bai and Yu-Buck, 2025). Negative user-generated content (UGC) is more significant than good UGC owing to negativity bias (She et al., 2024). Owing to the existence of negativity bias, some customers might give more weight to negative reviews (Wang et al., 2024). Owing to negativity bias, a negative event requires more information processing than a positive event when it is a necessity of understanding the causes of what happened (Söderlund, 2024). Having a negative bias can prevent retailers from pursuing other opportunities i.e., prior to starting a new venture, it is wise to assess the benefits and drawbacks, but delaying too long might cost retailers their competitive edge or prevent them from making the best possible launch into a new market. In addition, when it comes to customer feedback and brand reputation, negativity bias plays a significant role.

3. Research methodology

Empirical investigation of novel concepts or phenomena, with consequent theorization, has been propagated through qualitative analysis as a preferred method (Behera et al., 2024a). A qualitative approach facilitates interviews of key informants and captured their responses in their own words within the allotted times (Eisenhardt et al., 2016), which offers researchers new perspectives as opposed to just restating existing knowledge (Deshpande, 1983). To collect the most information from the interviews, this study used a semi-structured open-ended interview format, since it allows a deeper analysis and flow of the interviewee's opinions and knowledge (Madill, 2011). However, in the context of this research, the definition of suitable five- or seven-point Likert scale questionnaire surveys for the measurement items of metaverse-induced CBs and ISE was not available, and hence, a quantitative approach was not considered. In addition, when the subject of the study is not well understood and the objective is to generate propositions, qualitative approach is the best choice (Gros et al., 2024). Hence, qualitative analysis was considered as the research approach. To guarantee that the study is credible, dependable, and unique, the researchers used rigorous design to establish trust, ensure freedom of bias, and remain open to critique.

3.1. Study context

Data collection was done in India, which has been established as a developing country (Behera et al., 2023a). The rationale behind considering India as a data source is as follows: For decades, India's retail industry has been a major contributor to its growth and evolution (Ram et al., 2023). The diversity of the Indian retail industry and customer engagements has truly made it a force to be reckoned with metaverse. A growing interest in the utilisation and integration of immersive experiences, especially ISE, has been observed in India. For

example, luxury brands are using VR to offer virtual try-on of their products to enhance in-store ISE.

3.2. Respondents and retail firm profile

The key informants were brick-and-mortar retail store customers. Hundred informants were agreed to part of the study on the basis of their considerable experience with retail and the uses of metaverse. This was primarily done to avoid sampling bias. During the interview process, the interviewees and interviewers kept in touch via digital and interpersonal networks. The simple random sampling approach facilitates understanding of a subject or context that has to date been at best unclear (Teng et al., 2024). Hence, prospective informants were selected using a simple random sampling method for the personal interviews. Table 1 lists the participant's profile, which includes the retailer's size, the length of the corresponding interview, and their years of shopping experience.

The table contains profiles of six female and fourteen male customers who participated in the interviews. Firm sizes vary between [1-100], [100−1000], and more than 1000 in terms of employee strength, and the shopping experience in years varied between 0 and 5, 5–10, and more than 10 years. The average length of each interview was calculated to be 50 min. A total of 20 informants correctly provided response to the questions. For a qualitative analysis, a minimum sample size of 12 suffices for data saturation (Fugard and Potts, 2015). According to such a recommendation, a sample size of 20 is deemed sufficient for adequate data while simultaneously avoiding any form of bias. Therefore, this study achieved a saturation of 20, and further additions to the samples add little to no value.

3.3. Data collection procedure

Primary data were collected in two slots i.e., slot 1 (December 2023 to April 2024) and slot 2 (May 2024 to August 2024). Ten interviewers who were fluent in English and had rich endorsements conducted personal in-depth interviews. Such third-party interviewers were hired by the authors for this study and they were handed over twenty customer details for the collection of qualitative data. The interviews were conducted on a one-on-one basis over video conversation and discussion, which are subsequently recorded and transcribed by the interviewers. The interviewers fostered a casual atmosphere where participants felt free to express their opinions. In Slot 1, first round of 16 (serial no 1 to 16 in Table 1) in-depth interviews were conducted to build the base to seek answers whether CBs influence IS. In the process, the participants were informed about the study and its purpose to avoid any kind of bias in their responses. They were given overviews of the metaverse and its applications in retail, including its benefits, its relationship with innovation and improvement of customer service, the problems that can arise with customer involvement in an ISE and solutions to such problems, how to promote the concept of an ISE, CBs and how those biases affect their interactions in shopping, the occurrence and prevention of such biases, etc.

Table 1
Responder and brick-and-mortar retail firm profile (Author's own conceptualisation).

Sr No.	Gender	Firm size	Age	Industry type	Shopping experience	Interview length
1	Male	100–1000	32	Grocery	5–10	48
2	Male	>1000	55	Electronics	>10	49
3	Female	>1000	55	Clothing	>10	48
4	Male	0–100	31	Electronics	0–5	49
5	Female	100–1000	56	Grocery	>10	52
6	Female	0–100	50	Books	0–5	48
7	Male	0–100	40	Furniture	5–10	48
8	Male	>1000	57	Grocery	>10	53
9	Female	100–1000	53	Cosmetics	5–10	52
10	Male	0–100	32	Furniture	0–5	48
11	Male	100–1000	55	Art	>10	49
12	Male	100–1000	54	Furniture	5–10	50
13	Male	100–1000	37	Clothing	>10	51
14	Female	>1000	56	Cars	>10	52
15	Male	0–100	52	Grocery	5–10	46
16	Male	>1000	60	Books	>10	55
17	Male	100–1000	45	Sports	>10	55
18	Female	0–100	55	Cosmetic	0–5	50
19	Male	100–1000	50	Grocery	5–10	50
20	Male	100–1000	35	Grocery	5–10	55

Subsequently, a report of the level of adoption of metaverse tools and technologies in retail operational systems was asked based on the questions: (1) how much do they know about the use of metaverse technologies; (2) does the brick-and-mortar retail stores provide ISE; (3) does the brick-and-mortar retail store know about the ISE built on the metaverse; (4) how do CBs affect decision-making in immersive shopping? (5) what are the common types of CBs in immersive shopping; and (6) how the brick-and-mortar retail store reduce CBs? They were also asked to detail their experiences with CBs in their behaviour. After developing the initial understanding, the questions used in slot 1 were extended in slot 2 to understand the objectives, narratives, and roles of metaverse-induced CBs, negativity bias, and ISE.

Moreover, in Slot 2, the extended questions used to conduct 20 in-depth interviews were: (1) what the perception between metaverse-induced CBs and negativity bias is? (2) what the perception between negativity bias and ISE is? (3) how metaverse unfolds CBs; and (4) how metaverse shapes culture and society? The interviewers attempted to keep the interviews free-flowing while allowing the participants to explore and answer a broad spectrum of questions. Any follow-up questions were only asked if the interviewers required any further clarification on specific points about metaverse-induced CBs and their manifestations in their interactions with the metaverse. Fig. 1 summarises the data collection process by highlighting goals, semis-structured in-depth interview details, and the nature of the interview questions for Slot 1 and Slot 2.

3.4. Qualitative data analysis

Recording, transcription, analysis, and review of the data were constructed using the MAXQDA software tool, a comprehensive package for qualitative data analysis. A variety of qualitative data analysis approaches, such as content analysis, focus groups, case studies, and word cloud analysis, can be easily used using MAXQDA (Prasetyo, 2024). To retain the anonymity of the respondents and their firms, the names were omitted. The conduct of interviews and subsequent analysis were done alternatively by researchers. Instances emerged where applications and conclusions from data analysis were used to decide on additional requirements in the data. As per the recommendation by Behera et al. (Behera et al., 2023b), five discrete steps were followed as part of the qualitative metrology framework of the study: (1) data collection; (2) data organisation; (3) coding of the theory gathered; (4) data analysis for further insights; and (5) organising and reporting on the gathered conclusions since the primary objective of the research was to study and identify the qualitative patterns in the data through primary theoretical coding for thematic identification of categories, subcategories, and practices based on propositions formulated.

Following the interviewer's transcription of each interview, open, axial, and chosen coding was used to analyse the data to find an emergent theme that was consistent with inductive and deductive analysis (Azungah, 2018). Each investigator transcribed the interview and then submitted the results to MAXQDA. This helped in the 'open coding' process, which aimed to find recurring patterns in all of the interviews (Strauss and Corbin, 1990). The investigators can arrange, code, and analyse the data to highlight the emerging themes through MAXQDA's functionality. Following analysis, the themes that emerged from each interview were compared to see which ones were consistent, and those

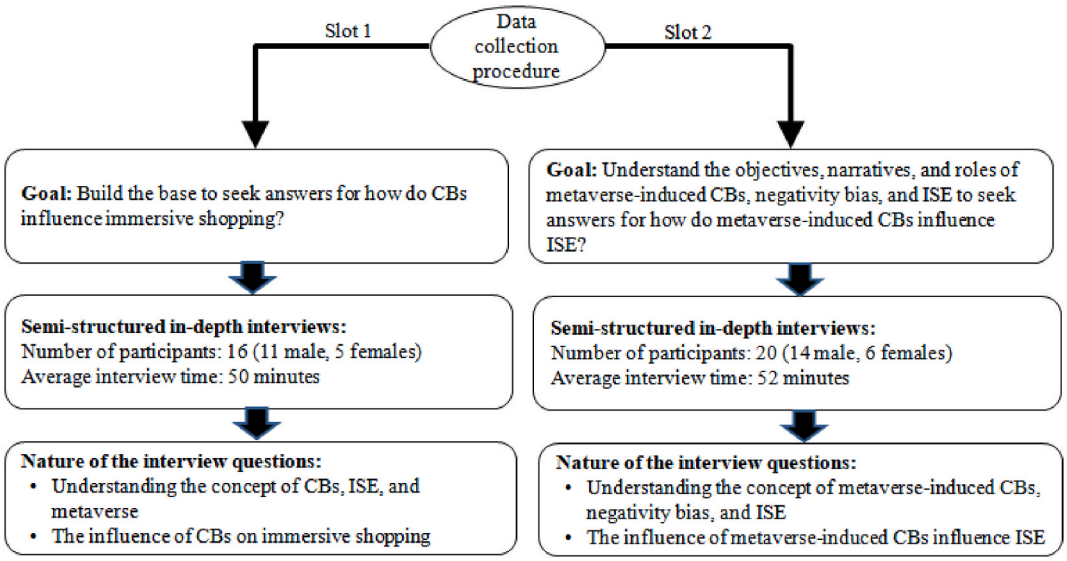


Fig. 1. Data collection process (Author's own conceptualisation).

results were then verified (Boyd et al., 2021).

This study followed Point & Fourboul's (Point et al., 2006) guidelines for coding, which stated that a coding unit is "an idea or a set of ideas that can be isolated from the rest of the qualitative data and presents certain coherence". The coding process identifies the themes by attaching labels. The data were analysed with an iterative process with back and forth coding process between data and successive interpretation. Until an agreement was reached, any discrepancies in the codes or coding guidelines were discussed by the investigators. This process allowed synthesising the empirical findings of the data (Gioia, 2021). Subsequently, as per the recommendation by Västhagen et al. (2022), the unit of analysis from quotation to theme comprising categories, subcategories, and an overarching theme was created (refer Table 2). In Table 2, N represents the number of customers who shared their opinion and percentage represents N % 20 (i.e., total number of informants).

4. Results

The results are structured as follows: First, it narrates the themes (refer Table 3) identified in the unit of analysis. Subsequently, it derives propositions from theme narrations.

Subsequently, the results are presented as a set of propositions from the theme and data analysis. The propositions are elaborated to establish the perceptions.

4.1. Perception between confirmation bias and negativity bias

Interview data revealed that customers can process metaverse-induced information in an irrational and biased manner based on the existing belief, which register negative stimuli to IS. It also revealed that such a bias can lead the customers to misjudge the products and services, resulting in dissatisfaction and poor decision-making on customer service to IS. Addressing it in the metaverse system can enhance their satisfaction. According to Zhao et al. (2020), extreme confirmation bias is the tendency for an individual to maintain opinions that are in line with their views while ignoring those that contradict them. When the confirmation bias pattern predominated, individuals' attitudes frequently reinforced (Sude et al., 2021). A homogeneous cluster forms as a result of decision-making specialists' confirmation bias, which causes them to interact primarily with individuals who possess similar views (Hu et al., 2025). To overcome this bias, the brick-and-mortar retail practitioners must enable their customers to investigate a variety of reliable and diverse sources of information in the metaverse, and simultaneously, take into account other viewpoints and scenarios. Hence, the proposition is postulated as:

P1: *For a brick-and-mortar retail customer, metaverse-induced confirmation bias stimulates negativity bias.*

4.2. Perception between availability bias and negativity bias

Interview data revealed that customers can choose to forego thorough analysis in favour of current or readily available metaverse-induced information when making decisions by overweighting new or recent information, which register negative stimuli to IS. It also revealed that such a bias can lead the customers to overestimation or underestimation that leaves them with low-quality information to form the basis of a decision to IS. Addressing it in the metaverse system can safeguard them against the fallacious reasoning that results in undesirable judgements. According to Cordero (2023), availability bias enhances emotion-induced uncertainty in the decision, and (Mamidala et al., 2023) can encourage individuals to believe erroneous concepts, stories, and historical events. Due to availability bias, people are more likely to be misled by information that is easily accessible, and when there is insufficient information available, they are more susceptible to negativity bias (Park and Kang, 2025). To overcome this bias, the brick-and-mortar retail practitioners must enable their customers to seek out diverse information sources from metaverse to assist them in combating the inclination to depend on readily available data. Whenever feasible, decisions should be made using metadata induced factual data rather than anecdotes or personal experiences. Hence, the proposition is postulated as:

P2: *For a brick-and-mortar retail customer, metaverse-induced availability bias stimulates negativity bias.*

4.3. Perception between anchoring bias and negativity bias

Interview data revealed that customers can skew their abilities to arrive at a logical conclusion, make accurate decisions, or select suitable choices due to existing belief in first-hand metaverse-induced information, which register negative stimuli to IS. It also revealed that such a bias can lead the customers to price and purchase anchoring to IS. Addressing it in the metaverse system can provide them with clear initial information on price and purchase and help to adjust away from initial anchors. According to Ziano and Villanova (2022), extreme

Table 2
Unit of analysis from quotation to theme (Author's own conceptualisation).

Category	N	Percentage	Sample quotation	Theme
Confirmation bias	16	80 %	"In my previously bad purchase from a particular brick-and-mortar retailer, I exhibit confirmation bias by actively seeking out fellow mates who have had similar experiences with either the same or different retailers' products or services".	**Theme 1:** Existing belief in IS information
Availability bias	17	85 %	"In my previously bad purchase from a particular brick-and-mortar retailer, I rethink my decisions to avail any type of products or services from the same retailer, despite evidence of their generally good quality of products or services at all times."	**Theme 2:** Overweight new or recent IS information
Anchoring bias	18	90 %	"For my successful purchase from a brick-and-mortar retail store at a specific price point, my inherent anchoring bias urges me to compare any existing version of the same or similar product's price to the successfully purchased product's price. If the price point of the newer version of the same product is greater than the anchored price, I decide to forego the purchase in favour of the previously bought product."	**Theme 3:** Existing belief in first-hand IS information
Egocentric bias	15	75 %	"When I encounter unsavoury information about a metaverse-based brick-and-mortar retailer's products or services, I draw my own conclusion about the same. When faced with contrary but true information about the purchased products or services, my egocentric bias leads me to deny it without regard for factual integrity."	**Theme 4:** Heavily belief on own IS information than reality
Hindsight bias	16	80 %	"When I encounter any information about a metaverse-based brick-and-mortar retailer's products or services, I draw my own conclusions about the same. In situations where I encounter any type of information that matches my assumptions about the retailer's products, services or decisions, my false feelings about conviction of having already predicted the actual outcome are reinforced, which can eventually lead me to make wrong and falsely informed decisions."	**Theme 5:** Existing belief on 'always knew' on IS information
Self-serving bias	17	85 %	"When I encounter unsavoury information about a metaverse-based brick-and-mortar retailer's PSDP, I draw my own conclusion about the same and act upon it. If the actions based on my convictions lead to a successful purchase, I credit the entire happening to my own judgement. If the actions result in a failed purchase, I refuse to acknowledge my part in the purchase and blame it entirely on the retailer and the metaverse platform."	**Theme 6:** Taking credit for IS information, regardless of the scenario
Choice-supportive bias	19	95 %	"When I have already drawn my own conclusions from rumoured information about a brick-and-mortar retailer's products or services, my successful purchase from a specific retailer gains more credibility. I proceed to urge other fellow customers to engage in business with that retailer while not only ignoring other retailer options but also falsely discrediting them against their products or services."	**Theme 7:** Tendency to belief on IS information choices as better than actuality
Conformity bias	18	90 %	"When I observe my fellow mates making generally successful purchases from a brick-and-mortar retailer's products or services, I tend to forego their individual decision,	**Theme 8:** Existing beliefs of others IS information without evaluation

(*continued on next page*)

Table 2 (*continued*)

Category	N	Percentage	Sample quotation	Theme
			products or services ability and attempt engagement with the popular retailer's products or services without validating their integrity or quality of products or services."	
Negativity bias	20	100 %	"When I already harbour or have started to harbour any form of negative feelings and assumptions about a metaverse-based brick-and-mortar retailer's products or services, I eventually dislike that retailer and metaverse."	**Theme 9:** Pay utmost attention to negative IS information
ISE	20	100 %	"My engagement with newer, more progressive technological integrations in a brick-and-mortar retailer's products or services makes me feel ahead of the curve and even urges me to encourage fellow mates to engage in the same. In the case of biases of adverse impact, I strongly discourage them, which is resulted in negative tone and poor attitude."	**Theme 10:** Dismantle deeper customer connection in IS

manifestations of the anchoring mechanism in customers eventually lead to unwillingness to purchase from a specific retailer or store. The anchoring mechanism in customers can also largely influence their dispositions on the metaverse negatively (Lanning et al., 2023). When creating decision environments, managers and employees should take anchoring bias into account since it remains prevalent across institutional contexts (Carter and Liu, 2025). To overcome this bias, the brick-and-mortar retail practitioners must enable their customers to collect more evidence-based information from metaverse and evaluate the reasoning. Moreover, multiple options can be considered to help in the decision-making. Hence, the proposition is postulated as:

P3: *For a brick-and-mortar retail customer, metaverse-induced anchoring bias stimulates negativity bias.*

4.4. Perception between egocentric bias and negativity bias

Interview data revealed that customers base their actions, ideas, and beliefs excessively on their own viewpoint due to heavily belief on metaverse-induced information than reality, which register negative stimuli to IS. It also revealed that such a bias can lead the customers to misunderstanding and potentially poor decision-making on product recommendations to others to IS. Addressing it in the metaverse system can influence their loyalty and retention. According to Sevi et al. (2022), customers with pre-conceived notions about online retail in the metaverse who are confident about said notions are either reinforced with greater intensity or lead them to completely invalidate any contradictory experience based on the precision of their notions. Overestimating

Table 3
Category with theme and its narration (Author's own conceptualisation).

Core category	Sub-category	Theme	Theme narrations
Metaverse-induced CB	Confirmation bias	Existing belief in IS information	• Brick-and-mortar retail customers that exhibit confirmation bias look for, read, and retains metaverse-induced information in a way that supports their pre-existing beliefs in IS. • Brick-and-mortar retail customers essentially hear what they want to hear, and regardless of how objective an individual may imagine themselves to be, they will favour metaverse-induced information that confirms what they already believe or want to be true.
	Availability bias	Overweight new or recent IS information	• When considering a purchasing choice in IS, the availability bias is a mental shortcut that relies on the metaverse-induced initial examples that come to a Brick-and-mortar retail customer's mind. • Brick-and-mortar retail customers incorrectly believe that their memories are an accurate representation of reality and base their decisions on what they remember. Because memories that are easily recalled are frequently insufficient for determining how probable events are to occur again in the future, the availability bias can result in poor decision-making and could have adverse effects on the general quality of life.
	Anchoring bias	Existing belief in first-hand IS information	• Anchoring bias occurs when brick-and-mortar retail customers give the first metaverse-induced information they encounter more weight than the second or third in IS. For instance, the first review someone sees about a product would have more of an impact than the second or third. • Brick-and-mortar retail customers' expectations are set by the first price they see for a retail product, and an individual can use this to influence how others react to product pricing.
	Egocentric bias	Heavily belief in own IS	• Brick-and-mortar retail customers exhibit egocentric bias when

(*continued on next page*)

Table 3 (*continued*)

Core category	Sub-category	Theme	Theme narrations
		information than reality	they place more metaverse-induced weight on their perspective of the IS than is actually true. Customers therefore undervalue or altogether disregard how dissimilar other individuals' ideas are from their own. • Egocentric bias can cause brick-and-mortar retail customers to ignore the needs and concerns of others while making decisions, often with unfavourable results.
	Hindsight bias	Existing belief in 'always knew' on IS information	• Brick-and-mortar retail customers exhibit hindsight bias who believe they can accurately predict how metaverse-induced event in IS would turn out in the future. The consumer may believe they should have known what would happen and that the anticipated issue was likely. • Decision-making is negatively impacted by hindsight bias which is rooted in overconfidence.
	Self-serving bias	Taking credit for IS information, regardless of the scenario	• Brick-and-mortar retail customers exhibit self-serving bias when it leads them to be overconfident, excessively take risks, overestimate their own abilities when on a winning streak, and vice versa based on metaverse-induced events in IS. More often, when they fail, they do not admit their own errors. • Self-serving bias has an impact on relationships, interpersonal dynamics, and self-esteem.
	Choice-supportive bias	Tendency to belief on IS information choices as better than actuality	• Brick-and-mortar retail customers' exhibit choice-supportive bias when it leads them to choose metaverse-induced recommendation for option A instead of option B to ignore or downplay the faults of option A while amplifying or subscribing new negative faults to option B. • Choice-supportive bias has a double-edged effect on Brick-and-mortar retail customers' ability to learn from their failures and make better decisions moving forward. It can give customers confidence in the decisions they make, but it can also inhibit them from doing so.
	Conformity bias	Existing beliefs of others IS information without evaluation	• Conformity bias is the tendency of Brick-and-mortar retail customers to behave in accordance with metaverse-induced information in IS rather than using their own personal judgment. • Conformity bias is driven by the desire of retail customers to blend in and prevent standing out. In the context of IS, this biasness explains why value-based pricing the default pricing approach is not used by the majority of retailers.
Negativity bias	None	Pay utmost attention to negative IS information	• Due to the negativity bias, Brick-and-mortar retail customers are more influenced by metaverse-induced negative events or information in IS than by positive counterparts of equal significance. • Negativity bias causes brick-and-mortar retail customers to dwell on the negative, making bad experiences seem much more important than they really are. This can then affect how customers make decisions and how they see other retail stakeholders.
ISE	None	Dismantle deeper customer connection in IS	• ISE helps brands stand out from their competitors. The Brick-and-mortar retail uses this to carve out a unique selling proposition (USP) and stay one step ahead of the competition. • The most useful feature of ISE is the shareability component, wherein immersive visual displays are ideal for brick-and-mortar retail customers to capture, record, and publish to social media. This kind of word-of-mouth promotion drives more customers to come into the store and also increases brand visibility online among potential customers who share mutual interests.

their ability to correctly conceptualise the benefits and disadvantages of specific products can largely distort their judgement on the same (Weiss, 2022). Egocentric bias has been considered an adverse indicator of explicit belief reasoning since it makes it difficult to reason about other people's beliefs (Speiger et al., 2025). To overcome this bias, the

brick-and-mortar retail practitioners must enable their customers to develop the awareness by exploring the alternative viewpoints from metaverse-induced information. Moreover, external input can be proactively asked to understand the feedback. Hence, the proposition is postulated as:

P4: *For a brick-and-mortar retail customer, metaverse-induced egocentric bias stimulates negativity bias.*

4.5. Perception between hindsight bias and negativity bias

Interview data revealed that customers consider an unforeseen incident in the past and conclude that it can be easily predicted due to existing belief in 'always knew' on metaverse-induced information, which register negative stimuli to IS. It also revealed that such a bias can lead the customers to create narratives that make sense of events in retrospect to IS. Addressing it in the metaverse system can reinforce their positive expectations and journey. According to Zwick et al. (1995), a customer's hindsight bias has a stronger influence on distortion of judgement and misconfiguration of satisfaction. If a customer already has or encounters unsavoury information about a retailer's products or services, their exaggeration of the already false information might be reinforced with a feeling of confirmation about the same (Geissmar et al., 2023). Foreseeability judgements are often plagued by hindsight bias (Güver and Kneer, 2025). To overcome this bias, the brick-and-mortar retail practitioners must enable their customers to consider how the metaverse-induced outcomes that did not unfold could have unfolded. Moreover, the metaverse can be designed to monitor customers past decisions and the associated predictions, which should be brought into their attention. Hence, the proposition is postulated as:

P5: *For a brick-and-mortar retail customer, metaverse-induced hindsight bias stimulates negativity bias.*

4.6. Perception between self-serving bias and negativity bias

Interview data revealed that customers blame external factors unconnected to their character for unfavourable outcomes and attach positive events and successes to their own acts or character due to the tendency of taking credit for metaverse-induced information, regardless of the scenario, which register negative stimuli to IS. It also revealed that such a bias can lead the customers to claim credit for good decisions and blame external factors (e.g., market conditions, competitors) for bad outcomes (e.g., if a product breaks or they receive poor customer service, they tend to blame the manufacturer or the store rather than their own actions or choices) to IS. Addressing it in the metaverse system can reinforce their self-compassionate, enhanced motivation and encourage resilience by fostering a more objective and balanced perspective. According to Liu et al. (2023), skewed perception can contribute to an overall negative view of online shopping in the metaverse, and also spread the said negative notion amongst other customers to strengthen their perceptions. Under the significant negative moderation effect of the result, the post-purchase relation between retailer and customer in the metaverse is found to be positive (Li and Li, 2021). Self-serving bias results in poor outcomes of business performance (Chen et al., 2025). To overcome this bias, the brick-and-mortar retail practitioners must enable their customers to assess the situation with metaverse by seeking feedback from others. Moreover, alternative viewpoints and expert opinions can be exercised with the interaction with metaverse. Hence, the proposition is postulated as:

P6: *For a brick-and-mortar retail customer, metaverse-induced self-serving bias stimulates negativity bias.*

4.7. Perception between choice-supportive bias and negativity bias

Interview data revealed that customers consider previous decisions are better suited than they actually were due to tendency in belief on metaverse-induced information choices as better than actuality, which register negative stimuli to IS. It also revealed that such a bias can lead the customers to justify purchases by defending their brand choices to IS. Addressing it in the metaverse system can reinforce their informed choices in the first place with unbiased information and feedback. According to Shuraida and Titah (2023), choice-supportive bias misremember past decisions in a more favourable light than warranted to avoid cognitive dissonance. Users are more prone to link the options with negative traits when they are subjected to choice-supportive bias (Siri et al., 2022). The choice-supportive bias leads to loss aversion (Watkins and Musselwhite, 2025). To overcome this bias, the brick-and-mortar retail practitioners must enable their customers to preserve objectivity while reflecting back their previous decisions by using structured decision-making, obtaining feedback and alternative viewpoints, or maintaining thorough archives of decisions to examine with the metaverse. Hence, the proposition is postulated as:

P7: *For a brick-and-mortar retail customer, metaverse-induced choice-supportive bias stimulates negativity bias.*

4.8. Perception between conformity bias and negativity bias

Interview data revealed that customers adapt to the beliefs and actions of those in their immediate vicinity due to existing beliefs of others metadata-induced information without evaluation, which register negative stimuli to IS. It also revealed that such a bias can lead the customers to social proof (i.e., individuals buy products or follow trends simply because many others do) to IS. Addressing it in the metaverse system can reinforce their sense of urgency. According to Padalia (2014), failure to identify such biases and act upon them can lead to misdiagnosis of actual issues in the retailer's products or services that require the customer's active involvement. The social tendency can manifest in the customer's decision to harbour a negative attitude towards a retailer's products or services simply because the majority of their fellow customers have fostered the same environment, since conformity bias can largely impact online group buying as well (Ou et al., 2022). Conformity bias hinders independent evaluations (Farahmand et al., 2025). To overcome this bias, the brick-and-mortar retail practitioners must enable their customers to actively seek out information from metaverse from the perspective of different background, experiences and expert opinions to make better decisions and develop self-awareness. Hence, the proposition is postulated as:

P8: *For a brick-and-mortar retail customer, metaverse-induced conformity bias stimulates negativity bias.*

4.9. Perception between negativity bias and immersive shopping experience

Interview data revealed that customers dwell negative stimuli more quickly, but also to reflect on these events due to their utmost attention to negative metaverse-induced information, which dismantle deeper customer connection in IS. It also revealed that such a bias can lead the customers to remember or focus more on negative experiences with IS by influencing their future purchasing decisions and brand perception. Addressing it in the metaverse system can highlight their positive aspects, encourage their positive reviews and critical thinking, and improve their balanced information that presents both positive and negative viewpoints. According to Akhtar et al. (2011), negative effects on a customer's mind-set can significantly influence their opinions and views on more progressive concepts and retail experiences. A common response to this situation might lead to the customer prying for even

more negative information to support her bias against the metaverse (Soroka et al., 2019). Negativity bias results in negative emotions (e.g., anxiety or stress) and causes a reverse effect on shopping attitudes (Li et al., 2025a). To overcome this bias, the brick-and-mortar retail practitioners must enable their customers to pay conscious attention on metaverse for positive experiences of IS over negative and reframe the positivity on enjoyment. Hence, the proposition is postulated as:

P9: *For a brick-and-mortar retail customer, negativity bias adversely stimulates immersive shopping experience.*

5. Discussion

Based on the propositions discussed in the results section, this study proposes a unique conceptual model (see Fig. 2), which applies the theoretical lens of metaverse-induced cognitive biases, negativity bias and the immersive shopping experience.

In the figure, propositions from P1 to P9 represent the perception effect. In fact, literature has proposed ISE-based conceptual models in the retail, i.e., Kim et al. (2023) analyse behavioural intention augmented with engaging ISE using VR technology and conceptualise how retailers can develop strong emotional relationships with customers and turn them into ardent supporters. The drawback is that the study was conducted for furniture retailers, and its findings cannot be generalised to other retail product types. In addition, existing studies have proposed conceptual models based on traditional shopping experiences (Barta et al., 2021) (Akhmaaj and Sharif, 2024) (Behera et al., 2024b) (Behera et al., 2024c) (Behera et al., 2024d), but those do not consider the key benefits of immersive experiences that the metaverse can transform the brick-and-mortar retail stores.

Such benefits include: (1) metaverse fosters undisturbed engagement between the customers and the products; (2) by coordinating artistic sensibilities with product slideshows, metaverse enhances the information retention of its customers; (3) by engaging with cues and heightening the sense of comfort and originality, metaverse encourages a distraction-free atmosphere; (4) customers can get self-assist solutions by overlaying virtual content over its real-world equivalent to simulate real-life events; (5) for convenience of access, metaverse offers multilingual support for customers who speak different languages; (6) to assure many customer touch points, metaverse can trigger contextual, event-based activities on several channels like email, short message service (SMS), and WhatsApp; (7) customers are given the next steps via metaverse, such as enrolling for a visual demo, or making a purchase and thus reducing or eliminating manual intervention; and (8) by placing the customer at the centre of the business process, metaverse helps in shifting gears.

Literature also discussed the benefits of immersive experiences in different settings; i.e., immersive experiences in the metaverse can prompt knowledge acquisitions and skill developments that lead to organisational benefits (Lee and Yu, 2025). Brand loyalty can be created with immersive experiences in the metaverse (Romero et al., 2025). Immersive experiences encourage players to interact with real-life destinations in the game industry (Zhu et al., 2025). Immersive experiences in the metaverse have the potential to enhance health and well-being outcomes and support the sustainable use of applications related to health (Shamim et al., 2025). An immersive experience is created to establish a high-quality organisational image (Wang et al., 2025). An immersive experience strengthens the cultural message's coherence (Zhang et al., 2025a). In AR, immersive experiences influence behaviour and satisfaction (Naveen et al., 2025). Immersive experiences have the ability to enhance student engagement and help make educators' learning more satisfying and memorable (Yuan and Liu, 2025).

Moreover, the rationale behind consideration of metaverse for ISE in the brick-and-mortar retail is as follows: The concept of the metaverse, a virtual environment where customers can communicate with one another and digital items, is quickly spreading around the world. The metaverse opens up new opportunities for growth and innovation in the retail industry, where it is expected to completely change how consumers purchase. In addition, the metaverse is already gaining popularity as a venue for ISE. A fundamental shift in how the retail industry views the customer experience has been brought about by the idea of IS in the metaverse. Brands are anticipating building immersive digital worlds that offer a more engaging purchasing experience rather than simply opening online storefronts, enabling customers to completely immerse themselves in the brand outside the physical store. The metaverse ISE combines elements of social networking and entertainment in retail to provide a whole experience. Brands now provide customers with swift access to product discovery, personalised recommendations, and frictionless shopping experiences because of the metaverse.

The study validates the perceptions about the impact of the metaverse and online user behaviour on ISE with previous studies, and the findings are consistent with those, i.e., augmented reality shopping behaviour requires immersion in metaverse environments (Zhang et al., 2025b). Businesses, academic institutions, and the government can jointly accelerate the development of novel ideas into practical products and facilitate virtual shopping in the metaverse across a range of industries (Zheng et al., 2024). The metaverse emphasises its intrinsically social aspect by enabling users to engage in virtual activities, like going to events or shopping with others (Kim et al., 2024). Streamers can synchronously present products from numerous views through live commerce in the metaverse, providing viewers with an engaging and dynamic shopping experience (Li et al., 2025b). The shopping

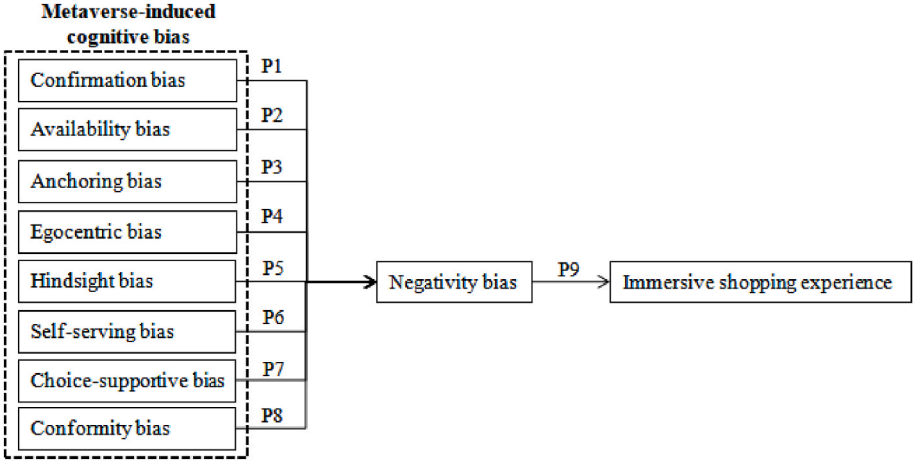


Fig. 2. Conceptual model (Author's own conceptualisation).

experience for online customers has changed as a result of the rise of metaverse (Pillai et al., 2025). By producing captivating consumer experiences during the shopping process and raising overall customer satisfaction, VR technology has the ability to completely transform the retail sector (Nim et al., 2025).

5.1. Qualitative findings

According to Glaser and Strauss (2017), while generating a theory from data, the majority of propositions and conceptions arise from the data itself and are methodically developed in connection to the data throughout the qualitative research process. Therefore, the qualitative findings are the reflection from author's conceptualisation of the theoretical underpinnings. However, in addition to statistical generalization, there can be multiple forms of generalizability of the findings (Yin, 2009) (Lee and Baskerville, 2003), and hence, the generalizability of the qualitative findings cannot be an exception. Therefore, this study aims to generalize the findings from statements made by the respondents. According to Lee and Baskerville (2003), the generalization of the findings is not being proven, but well-formed proposition's derivatives.

The findings lead to three potholes owing to metaverse-induced CBs. First is the digital divide in the ISE by corroborating propositions P2, P6, and P9. Digital divide refers to the gap that exists between customer communities that have levels of ownership or access to the metaverse, and those communities that do not. Due to metaverse-induced CBs, the integration of retail services with the metaverse aggravates the risk of increasing the gap. A key factor in the effective application of the metaverse in the retail service ecosystem is the dynamics influencing how customers interact with their communities and how much they trust the metaverse. However, the metaverse revolution has changed how customers from different cultures operate on a global, regional, and individual level, and as a result, the ISE has evolved. Therefore, measures by retail businesses must be considered for mitigating metaverse-induced CBs in IS by (1) applying principles, including removing sensitive variables such as race, gender, or age, as these can easily introduce biases during analysis; (2) using bias mitigation techniques such as bias amplification and biased sampling; and (3) using human oversight and decision-making to guarantee that any digital biases have been identified and rectified. This finding is in line with the prior study by Papadopoulos and Cleveland (2023), i.e., the digital divide relates to culture.

The second is financial malfeasance by corroborating propositions P4, P5, P8, and P9. It refers to a significant accusation that can be made against directors or executives of retail firms who are in financial trouble and may be found guilty of failing to uphold their fiduciary duty. The metaverse-induced CBs make it susceptible to financial incidents, such as the exploitation of codes and the illegal establishment of online shops and retail services that offer customers seemingly fool proof service while also looking at their financial data and exploiting its benefits. Such incidents can further encourage the misconception that the metaverse is an unsafe environment for mainstream activities like IS. Retail firms may face major variations in profit and loss when dealing with such an unsafe financial environment. Revenue generation for any associated digital stores is in danger of being manipulated during such unwarranted incidents. In addition, an internal attack on an online shop's financial database holds grave consequences for both the firm and the metaverse, which guarantees a growing sense of mistrust. Therefore, retail businesses must mitigate metaverse-induced CBs in IS by establishing effective policies to zero in on problems with personnel recruitment, organisational realignment, and training. This finding is in line with the prior study by Pham (2020), i.e., executives with legal expertise must be aware of inappropriate behaviour spurred by financial malfeasance.

The third is the business reputation risk, including adversely impacting decision-making, strategy formulations, and outcomes with serious problems including loss of customer trust, compromised brand reputation, increased regulatory scrutiny, loss of employee trust, and eroding shareholder value, by corroborating propositions P1, P3, P7, and P9. With the possible occurrence of such unwanted biases, the reputation of retail businesses and the metaverse can go down significantly. Reputation can be hard to improve and even harder to maintain on newer and more up-and-coming platforms. As the business reputation declines, the probability of the very same effect taking place in the immersive shopping experience increases and raises the risk of eventual distrust from customers. Therefore, to mitigate metaverse-induced CBs in IS, measures by retail businesses must be considered in the metaverse to safeguard their reputation, including (1) testing algorithms in a real-life setting; (2) accounting for counterfactual fairness; (3) considering human-in-the-loop systems; and (4) changing the way people are educated about marketing and technology. However, such measures would be beneficial, but some issues can call for a multidisciplinary strategy in addition to metaverse technology solutions. This finding is in line with the prior study by Wang et al. (2021), i.e., customers pay attention to business reputation.

5.2. Theoretical contributions

The study encapsulates radical contributions to theory. First, the study proposed a unique conceptual model with metaverse-induced CBs as the independent variables, negativity bias as an intermediate variable, and ISE as the dependent variable. As per Lin et al. (2023), there is a negative correlation between CBs and purchase intention (PI). As per Yang and He (2011), there is a positive correlation between PI and customer experience. By extending the extant research in the area of the CBs, this study establishes the perceptions between metaverse-induced CBs, negativity bias, and ISE. Therefore, it pushes the literature of CBs to metaverse and ISE. With the rigorous data analysis, this study provides a strong evidence for the existence of CBs in retail, which is commonly recognised as one of the primary causes of market inefficiency and has the potential to exacerbate equity mispricing (Sun et al., 2024). Moreover, the negativity bias can affect purchasing decisions and brand judgement in IS. This provides strong evidence to Acciarini et al. (2021), which states that decision-making is influenced by CBs. Therefore, this contribution adds a new metaphor to the field of immersive customer experience in brick-and-mortar retail, with consideration for mitigating metaverse-induced CBs. Therefore, brick-and-mortar retailers must continuously monitor and evaluate metaverse-induced CBs by integrating metaverse-induced CBs-detection algorithms for rectification.

Second, ISE is a virtual shopping centre that might allow customers to explore the environment freely while navigating the aisles, browsing through offered products, and engaging with promotional media, and often times, the support schemes that offer customer assistance (Hsu et al., 2024). Retail 4.0 is a recently coined term for the industrial improvement in retail that involves AI, Cloud Computing, Big Data Analytics, and AR/VR to enhance the already existing concept of Retail 3.0, which offers customers online shopping experiences. Integration of technology, innovation, and people make Retail 4.0 a smart concept, and ISE fits right into its implementations (Kamble et al., 2020). In Retail 4.0, the use cases for consideration of metaverse for ISE includes virtual try-on that allows customers to try on makeup, clothing, and other products before they buy them and by utilising AR, VR and AI, the metaverse replicates the in-store fitting experience. Hence, metaverse-induced CBs and negativity bias in particular may contribute to social inequalities in ISE. In this vein, this contribution forged a link between Retail 4.0 and mitigating metaverse-induced CBs in ISE. Therefore, brick-and-mortar retailers must focus on regular audits, transparency, and accountability to adhere to Retail 4.0 regulations.

5.3. Managerial implications

Brick-and-mortar retail managers can find the study to be highly beneficial in the context of remedying metaverse-induced CBs for an ISE. First, in the dynamic realm of business, customer awareness and

interaction is crucial in determining the success and the reputation of the organisation. So, for the better operationalizations of the business reputation, the managers of a brick-and-mortar retail can make customers aware of the fallacies that can take shape in the metaverse, which is a crucial step to increasing trust and involvement. So, regular interactions about their experiences and receiving feedback about metaverse-induced CBs can serve as a guiding resource to adjust and adapt the customer's ISE over time. This may have a favourable effect on decision-making, which minimises unfairness or inconsistent treatment of customers in IS. However, if one customer group receives preferential treatment over another group in IS, as is frequently the case when handling complaints, customer dissatisfaction can be rife. As per Nouri-Harzvili and Hosseini-Motlagh (2023), customer dissatisfaction results in revenue loss. So, the managers of a brick-and-mortar retail must mitigate the metaverse-induced CBs in IS by quickly seizing the opportunity to address the issue with such customer groups by analysing the performance of the metaverse to identify the root causes.

Second, with the diversity of stakeholders, metaverse-induced CBs are less likely to develop. So, for the better operationalisations of the stakeholders involvement for the success of the business, the managers of a brick-and-mortar retailer can surround an increasing number of subject-matter experts, developers, technology firms, and organisational diversity to address metaverse-induced CBs that can encourage customers to experience IS in a safer, more comfortable, and known environment. So, before addressing metaverse-induced CBs, stakeholders must understand how they affect the business by conducting audits, establishing processes, implementing policies, and observing the outcome. Subsequently, managers can look for patterns, gaps, and disparities that may indicate a CB. Thereon, stakeholders must implement best practices for promoting fairness and transparency in the metaverse to mitigate the impact of CB or eliminate it. However, addressing metaverse-induced CB is not a one-time event but an ongoing effort. Therefore, managers must mitigate the metaverse-induced CBs by promoting a culture that honours and respects diversity, conversation, and feedback and fosters learning and development for the digital maturity of IS. As per Pinto et al. (2023), the road to digital maturity is grounded in technology and culture.

Third, continuous governance efforts in the metaverse can enable one to quickly identify and respond to compliance and risk issues as they arise. So, for the effective handling of the compliance and risk issues, the managers of brick-and-mortar retailers can regularly monitor customer data to address metaverse-induced CBs and bring about a safe and trustworthy environment for ISE. However, metaverse-induced CBs can harm engagement and wellbeing by clouding customers' sense of objectivity and fairness in the metaverse. In addition, it can have an effect on day-to-day operations, including security and poor business decisions, as well as lost customer trust and significant legal and ethical exposure. So, managers must mitigate the metaverse-induced CBs in IS by recognising the importance of addressing and reducing such CBs to promote fairness and inclusivity. Therefore, showing customers that their feedback and opinions are being taken into account while governing the metaverse helps managers create a sense of inclusivity. Consequently, it also serves as an effort to regularly supervise and update the environment for digital strategy. As per Pinto et al. (2023), a holistic management approach for the digital strategy is associated with governance.

5.4. Limitations and future research directions

There are some limitations, which consequently open new doors for further research. First, the objective of this study is to present a theory-building approach rather than a comprehensive empirical analysis. As a result, any conclusive decisions about metaverse-induced CBs in ISE must be taken with great caution and judgement. Henceforth, further research can be propagated through in-depth surveys and quantitative assessments for the validation and evolution of the proposed conceptual model and its theoretical underpinning with practical implementations. Second, it is important to note that the large-scale implementation of the metaverse-induced CBs solutions might not be possible due to a lack of intellectual understanding. Therefore, experienced and knowledgeable resources are crucial in implementing the remediation of problems arising due to metaverse-induced CBs in consumerism. However, improper or inefficient remedies can cause an enormous financial drain on businesses involving ISE. Henceforth, further research can be conducted to reproduce metaverse-induced CBs in the context of ISE for the other retail settings.

Third, this study considers CBs that are induced by IT, like the metaverse. However, other biases, such as algorithmic bias, the halo effect (Maldifassi and Canessa, 2009), Gambler's fallacy (Stöckl et al., 2015), and the Dunning-Kruger effect (Gignac, 2024), were not taken into account since the scope of the study was confined to CBs. Therefore, a future study can be conducted to explore such biases and their perception with negativity bias by adopting a qualitative approach in the context of ISE. Fourth, the bulk of the data were collected from India, a developing nation where the metaverse has not yet been established as the most common and promising technology in ISE for brick-and-mortar retail. Therefore, the future study can be conducted in a developed nation where the metaverse is considered to be the backbone of ISE, which might provide insights that are more intricate and explanatory.

Fifth, the study has adopted qualitative research with a sample size of 20, which is a relatively small number in comparison to the sample size considered in quantitative research. However, many qualitative researchers claim that sample size is not an issue in qualitative research because the majority of qualitative research does not entail drawing statistical generalisations (Tutar et al., 2024). Moreover, a small sample size in qualitative research may yield results that may not be representative of the population, but a large sample may need more time and resources to analyse, thereby resulting in a superficial analysis and decreased depth. Sometimes, a small sample makes it difficult to generalize the findings to a wider audience and can lead to a narrow and incomplete understanding of the CBs in ISE. Hence, statistical generalizability of the propositions to a broader population is required for a broad and complete understanding of CBs in ISE. Therefore, future studies can be conducted to determine the Likert scale measurement instruments of the proposed biases, ISE, and establish a structural relationship between them with the help of a quantitative approach for statistical generalisations of the findings.

6. Conclusions

The study proposed a unique conceptual model demonstrating the adverse impact of cognitive biases on the immersive shopping experience for brick-and-mortar retail. A qualitative approach has been used for the empirical analysis, and this study produced three findings. First, metaverse-induced CBs cause a digital divide in human culture. Second, metaverse-induced CBs cause financial malfeasance that may result in a financial scandal harming stockholders. Third, metaverse-induced CBs cause business reputation risk that can devastate business operations, with possible worst-case scenarios adversely impacting revenues as well as firm positions. When compared to a traditional shopping experience, immersive shopping experiences offer brick-and-mortar retailers distinct advantages, including acceleration of the decision to buy, boost conversion rates, lower the number of returns, and foster more consumer loyalty. Considering the importance of immersive shopping experiences in this digital era, brick-and-mortar retailers must adopt a culture in which metaverse-induced CBs should be identified, acted upon, and prevented for the greater good of the organisation.

CRediT authorship contribution statement

Soumili Ghosh: Writing – original draft, Methodology, Investigation, Formal analysis. **Rajat Kumar Behera:** Writing – review & editing,

Writing – original draft, Conceptualization. **Pradip Kumar Bala:** Supervision, Formal analysis. **Nripendra P. Rana:** Validation, Supervision, Conceptualization.

Data availability

Data will be made available on request.

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